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Review

A taxonomy of line balancing problems and their solution approaches

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A B S T R A C T

Line balancing belongs to a class of intensively studied combinatorial optimization problems known to be NP-hard in general. For several decades, the core problem originally introduced for manual assembly has been extended to suit robotic, machining and disassembly contexts. However, despite various industrial environments and line configurations, often quite similar or even identical mathematical models have been developed. The objective of this survey is to analyze recent research on balancing flow lines within many different industrial contexts in order to classify and compare the means for input data modelling, constraints and objective functions used. This survey covers about 300 studies on line balancing problems. Particular attention is paid to recent publications that have appeared in 2007–2012 to focus on new advances in the state-of-the-art.

Contents

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1. Introduction

A flow line consists of a sequence of workstations performing repetitive sets of tasks on products. Such lines are used in many manufacturing contexts such as machining, assembly, or disassembly. Because of the high investment and running costs involved, the design (or re-design) of such lines is of considerable practical importance (Askin and Standridge, 1993; Nof et al., 1997; Scholl, 1999; Dolgui and Proth, 2010). A number of crucial decisions have to be made in flow line design, including product design, process selection, line layout configuration and line balancing. Usually these problems are considered one at a time because of their complexity (Kimms, 2000; Zhang et al., 2002; Battaïa et al., 2012a).

The first two steps – product design and process selection – provide the information about the work that must be done in the flow line being designed, i.e., a set of indivisible tasks related by some constraints. The sources of these constraints can be the technology used, economic and environmental considerations or ergonomic factors for the workforce. The next step deals with the choice of the line layout (straight, U-shaped, with circular transfer, asymmetric, etc.). This defines how the workstations will be situated on the line as well as what flow directions and rules are to be used. Finally, the last and crucial step is line balancing. Here tasks are assigned to the workstations and resources that will be deployed on the line. This is a complex combinatorial problem. Its solution determines for the most part the efficiency of the line designed.

This paper surveys the contemporary research literature on line balancing and presents a taxonomy for the wide range of models and solution approaches proposed for these problems to enable researchers and practitioners to understand better the current state of this domain. The goal is to help to find the nearest known models and possible methods for future line balancing problems. Unlike the majority of other state of the art studies, our study is not limited to assembly lines but considers diverse line balancing problems from different contexts – machining, assembly, disassembly and other industries – within a common modelling framework.

This article emphasises the topics being intensively studied for the last decade. More than 267 papers have been published in major refereed international journals since the last comprehensive review (Boysen et al., 2008) became available online in 2006. This significant amount of publications shows that this subject continues to hold an important place in production research. The current article is especially focused on these most recent publications. Nevertheless, it highlights also key issues identified in the past but which have unfortunately remained undeveloped and still require additional academic research.

The first known formulation of an assembly line balancing problem has been made by Salveson (1955). It assigned a set of tasks $I=\{1,2,\ldots,i,\ldots,|I|\}$ to linearly ordered workstations $M=\{1,2,\ldots,k,\ldots,m\}$. Order relations among the tasks are given by a precedence graph G, where an arc (i,j) exists if task j cannot be started before the end of task i. The tasks assigned to workstation k, i.e., set I_k, are performed sequentially, i.e., workstation processing time, $T_k=\sum_{i\in I_k} t_i$, where t_i is the processing time of task i. The cycle time constraint requires that workstation processing times do not exceed a given value c referred to as line cycle time (also known as takt time), i.e., $T_k\le c$, $\forall k \in M$. The objective is to assign all given tasks with respect to precedence and cycle time constraints while minimizing the number of workstations required. This problem was referred to by Baybars (1986) as the Simple Assembly Line Balancing Problem (SALBP). SALBP is known to be NP-hard in general (Wee and Magazine, 1986). Several indicators have been suggested to evaluate the computational complexity of particular SALBP instances (for more details, see Bhattacharjee and Sahu, 1990; Driscoll and Thilakawardana, 2001; Hoffmann, 1990; Scholl, 1999).

SALBP has been intensively studied in the literature. As a result, numerous operational research techniques have been applied to solve this problem to optimality or approximately. Several evaluations of heuristics (Boctor, 1995; Ponnambalam et al., 1999; Talbot et al., 1986) and exact methods (Baybars, 1986; Erel and Sarin, 1998; Scholl, 1999; Scholl and Becker, 2006) have been presented. However, a number of recent publications show that SALBP is still a challenging topic for researchers (e.g., Bautista and Pereira, 2009; Blum, 2008; Ho and Emrouznejad, 2009; Kilincci, 2010, 2011; Kilincci and Bayhan, 2008; Liu et al., 2008; Nearchou, 2007; Özcan and Toklu, 2009a; Pastor and Ferrer, 2009; Sewell and Jacobson, 2012; Sheu and Chen, 2008).

Recently, an algorithm called "Branch, Bound, and Remember" was presented by Sewell and Jacobson (2012). It provided optimal solutions for 269 test-bed instances referenced in the literature in less than half a second per problem, on average. This is an excellent result. Nevertheless, this does not signify that the research on SALBP is now definitively over. It is known that for combinatorial problems there exist always examples for which enumeration algorithms are not efficient enough. Thus, the search for new and more difficult SALBP benchmarks is an interesting path for future research.

The SALBP considers a single straight assembly line for only one type of product. This problem was generalized for other line configurations, manufacturing contexts and performance requirements. Each novel formulation involved new decisions, constraints or/and optimization objectives. As a result, a great number of definitions have appeared for the so-called Generalized Assembly Line Balancing Problem (GALBP), Disassembly Line Balancing Problem (DLBP) and Transfer Line Balancing Problem (TLBP). However, similarities can be observed amongst them, e.g., both assembly and disassembly lines can have a U-layout, multiple workplaces or employ workers with different skills. Precedence constraints with standard and exclusive (OR-) relations can be appropriate for machining, assembly, and disassembly processes. As a consequence, when only looking at the mathematical model of a line balancing problem, often one cannot identify if the model has been developed for assembly, disassembly or machining line. Thus, our survey aims to present a novel taxonomy, more general and common for several manufacturing environments where flow lines are used.

Taking into account the ever growing number of articles on generalisations of SALBP, the previous comprehensive surveys on GALBP (Ghosh and Gagnon, 1989; Gagnon and Ghosh, 1991; Rekiek et al., 2002; Becker and Scholl, 2006), while still valuable in themselves have not covered the numerous current issues and the large body of recently published research.

The survey of Boysen et al. (2007) has proposed the first and very interesting classification of line balancing problems in the form of the tuple-notation. The tuples corresponded to:

(i) precedence graph characteristics, (ii) station and line characteristics and (iii) objectives. About 150 papers published between 1966 and 2006 were evaluated in those surveys. This approach of line balancing classification has also been utilised in Boysen et al. (2008).

Other recent reviews have mainly focused on specific topics, like multiple assembly lines (Lusa, 2008), applications of genetic algorithms (Tasan and Tunali, 2008) or self-balancing production lines (Bratcu and Dolgui, 2005). Thus, they have not addressed the entire scope of line balancing research.

None of the aforementioned surveys have considered such present trends in the literature as balancing disassembly lines, problems with fuzzy or interval-given task attributes, ergonomic or environmental constraints. Moreover, multiple-criteria decision making has not been adequately treated when compared with its place in today's literature. All of these issues are discussed in the current survey. It is not limited, as previous ones, to assembly lines. It seeks to present a larger perspective including line balancing problems in machining and disassembly. The objective is to develop a novel taxonomy in order to analyze recent studies published after 2006, and to identify the principal weaknesses and undeveloped points in the current research. To clarify these points, some references before 2007 will be also given. This taxonomy is based on the following five elements, which will be addressed in Sections 3–7, respectively:

(i) Number of lines to be balanced;
(ii) Task attributes $T(i)$;
(iii) Workstation attributes $W(k)$;
(iv) Constraints to be respected by a feasible solution;
(v) Criteria used to distinguish better or when possible the best (optimal) solutions.

The first three elements have a significant impact on the number of decision variables whose values must be determined. They give an idea on the problem's size and structure. The last two elements, (iv) and (v), determine principally the type of the problem to be solved, whether linear or nonlinear, as well as whether single or multi-objective.

The rest of the paper is organized as follows. Section 2 introduces the principal characteristics of flow lines. Sections 3–7 consider, respectively multiple lines, task attributes, workstation attributes, constraints and optimization criteria. Section 8 presents an overview of existing solution methods. Section 9 considers in detail multi-criteria decision making in the line balancing context. Section 10 discusses the current trends and perspectives of present research on line balancing. Concluding remarks are given in Section 11.

2. Line balancing: General principles

Since various manufacturing features of flow lines have been frequently discussed in the literature of line balancing (Askin and Standridge, 1993; Battaïa et al., 2012a; Boysen et al., 2007, 2008; Dolgui and Proth, 2010; McGovern and Gupta, 2011; Nof et al., 1997; Scholl, 1999), only brief descriptions will be provided here for readers not familiar with this topic.

2.1. Industrial environments

The principal industrial environments for which line balancing problems are considered are:

– Machining: a part is completed by a series of machining operations like drilling, milling, reaming, etc. In general, there may be much fewer precedence relations between such operations than in the case of an assembly/disassembly process. However, there may exist a lot of tolerance constraints that impose assigning operations to the same workstation and/or incompatibility constraints that forbid the assignment of certain operations to the same workstation because of technological incompatibilities. Such lines are usually highly automated. For more details, see Battaïa et al. (2012a), Delorme et al. (2009), Dolgui et al. (2008a, 2009a), and Guschinskaya and Dolgui (2009).
– Assembly: a final product is obtained by assembling a number of components. The precedence graph may have many initial nodes, but usually results in a single final node. These lines can have various configurations from manual with workers assigned to each workstation to completely automated. For more details, see Boysen et al. (2007, 2008), Rekiek and Delchambre (2006), and Rekiek et al. (2002).
– Disassembly: a number of parts or sub-assemblies are obtained from an initial product. Generally, the precedence graph cannot be derived by reversing the initial assembly precedence graph. Moreover, the final state is not always predetermined and because of this, such lines are mostly manual today. For more details see (Ilgin and Gupta, 2010) and (McGovern and Gupta, 2011).

Fig. 1 presents possible precedence graphs for machining, assembly, and disassembly lines, respectively.

2.2. Number of product models

According to this criterion, the following line types are commonly distinguished in the literature:

– Single-model lines: one homogeneous product is manufactured at the line. A set of all tasks to be performed for a product item is known. Each workstation performs a subset of these tasks. This subset is the same for all cycles. Therefore, the given set of all tasks must be partitioned among the workstations of the line and only one subset is associated with each workstation. For more details, see Dolgui and Proth (2010), Dou et al. (2011), Kara et al. (2009).
– Mixed-model lines: several models from a basic product family are manufactured simultaneously. The main processes for the models are quite similar, since they differ from the basic product with respect only to some attributes and

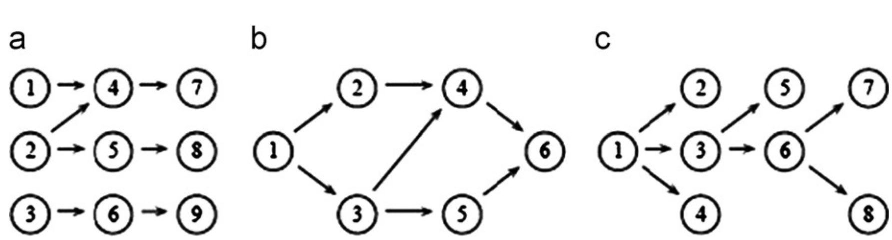

Fig. 1. Precedence graphs.

optional features. Therefore, a given set of all tasks must be partitioned among the workstations by associating a subset of each model with each workstation. Thus, the number of different subsets associated with each workstation corresponds to the number of models produced. For more details, see Battini et al. (2008, 2010), Boysen et al. (2009), Emde et al. (2010), Kara et al. (2011a), Tonelli et al. (2012), Yang et al. (2011).

– Multi-model lines: several products are manufactured in separate batches. In this case, the line can be rebalanced for each batch. Setup times need to be considered between the lots. For more details, see van Zante-de Fokkert and de Kok (1997), Boysen et al. (2008).

2.3. Line layout

Line layout defines the rules for task processing at workstations. In line balancing, these rules are taken into account principally in the form of problem constraints, as presented in detail in Section 6. The following line layouts are often considered in the literature.

– *Basic straight lines*. Each workpiece visits a series of workstations in the order of their installation as shown in Fig. 2. A set of tasks is assigned to each workstation. The tasks are executed one after the other. For more details see Kara et al. (2009, 2011b), and Gökçen et al. (2010), for example.
– *Straight lines with multiple workplaces*. Workstations are aligned as shown in Fig. 2. However, at each workstation, a number of parallel workplaces (Battaïa et al., 2012b; Becker and Scholl, 2009; Borisovsky et al., 2012a; Chang and Chang, 2010; Delorme et al., 2012; Kahan et al., 2009), serial workplaces (Battaïa et al., 2012c; Battaïa and Dolgui, 2012; Dolgui et al., 2008c; Guschinskaya et al., 2008) or mixed activated workplaces (Dolgui and Ihnatsenka, 2009a) are installed in such a way that the workers or pieces of equipment associated with each workplace can perform simultaneously, sequentially or in a series-parallel way on each workpiece, respectively. These three cases are shown in Fig. 3.
– *U-shaped lines*. These lines have both the entrance and exit in the same place. They are commonly manual. Workers placed between two legs of the line are allowed to walk from one leg to another as shown in Fig. 4. Therefore, they can work on two (or more) workpieces during the same cycle (Ağpak and Gökçen, 2007; Bagher et al., 2011; Hwang and Katayama, 2009; Hwang et al., 2008; Kara et al., 2007, 2011b; Sabuncuoglu et al., 2009; Toklu and Özcan, 2008; Toksarı et al., 2008). In this case, several subsets of tasks associated with different workstations are performed by the same worker. In the mathematical models of these lines, precedence and cycle time constraints are not treated in the same manner as for straight lines.
– *Lines with a circular transfer*. Workstations are installed around a rotating table (see Fig. 5) which is used for loading–unloading and moving the part from a workstation to another. With regard to the number of turns during which a part stays on the table before being completed, the lines with single (Battaïa et al., 2012a; Dolgui et al., 2008c) and multi-turn (Battini et al., 2007) circular transfer can be distinguished. If only one part side is treated at each workstation and a single turn is sufficient for completing a product, then this configuration is equivalent to a basic straight line. If several sides of the part can be treated simultaneously, then this configuration is equivalent to a line with multiple parallel workplaces. For the case of multi-turn transfer, the set of tasks assigned to a workstation must be partitioned into the different cycles corresponding to the number of turns of the table.
– *Asymmetric lines*. Asymmetric line configurations have been considered by AlGeddawy and ElMaraghy (2010), Ko and Hu (2008), and Li et al. (2011). They can be used to postpone the differentiation of products in order to maintain a common line configuration for all manufactured products as long as possible. This strategy reduces the risks associated with increasing product variety, but the corresponding line balancing problem must be solved conjointly with a layout optimization problem in order to determine the final line configuration. The layout with the main and secondary feeder lines can be also cited here (Azzi et al., 2012a, 2012b).

Any combination of the above layouts may exist when parallel lines are employed. The possible layouts for combining multiple lines are examined in the next section.

3. Multiple lines

Using multiple lines can offer a number of advantages. Investments can be deferred, because additional lines can be installed one by one as they are needed. Production can be adapted more easily to meet changes in demand. Failure of one line does not necessarily adversely affect the rhythm of other lines. Production cost may be reduced, etc.

One drawback is the increased investment cost compared to a single line. The consequences on worker productivity can be diverse. A longer line cycle time in the case of multiple lines enriches the work and improves motivation. However, this may limit the learning effect, since the variation of operations performed by a worker increases. Therefore, the choice between a single line with a short cycle time and a number of lines with longer cycle times is not straightforward and should be considered as a part of an integrated decision making process.

The survey presented in Lusa (2008) lists the possible configurations of multiple lines:

- Lines are independent with identical or different configurations (Chiang et al., 2007; Pinnoi and Wilhelm, 1997b).
- Lines with multi-line workstations or workers performing tasks for more than one line (Gökçen et al., 2010; Ozbakir et al., 2011; Özcan et al., 2010; Rabbani et al., 2012; Scholl and Boysen, 2009). Installing multi-line workstations may result in

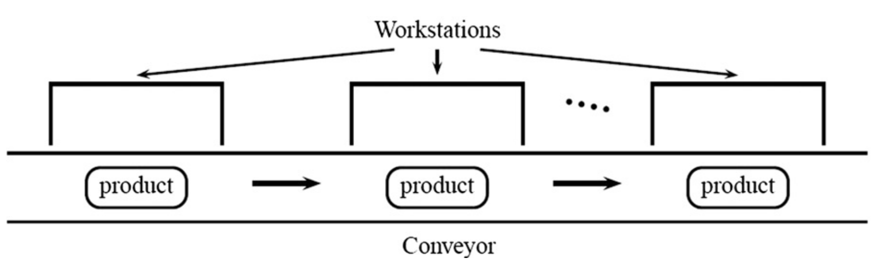

Fig. 2. A basic straight line.

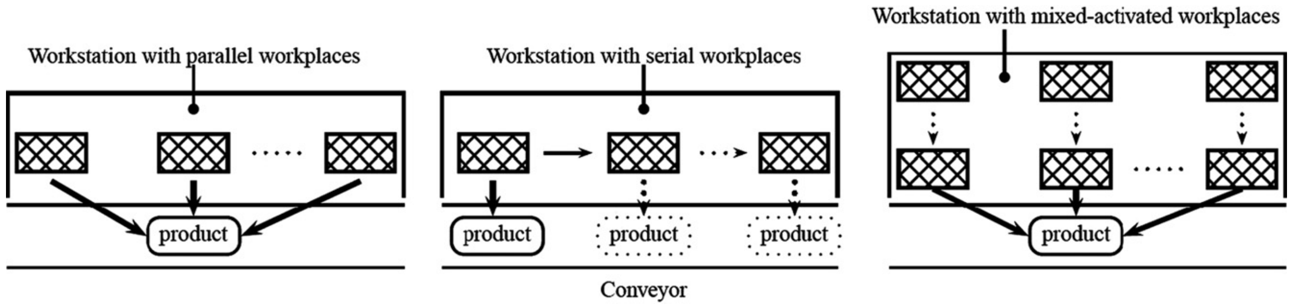

Fig. 3. Straight lines with multiple workplaces.

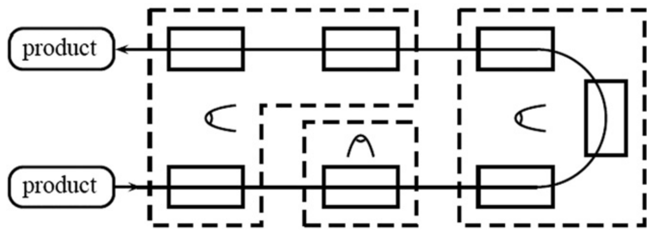

Fig. 4. U-line.

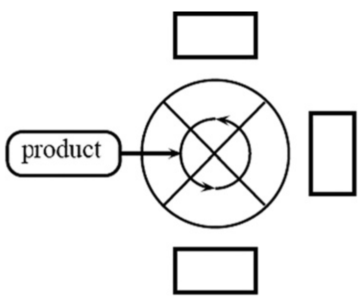

Fig. 5. Production system with circular transfer.

a reduction in the total number of workstations required, but any breakdown of a multi-line workstation may affect the production of more than one line.

- Parallel lines with crossovers between some workstations, where such crossovers can transfer workpieces from one line to another in the event of failures (Spicer et al., 2002).

In addition to the above mentioned cases, designing multiple lines is also required when a main line is connected to secondary feeder lines (Azzi et al., 2012a, 2012b).

The consideration of multiple lines increases dramatically the complexity of the line balancing problem, since it is necessary to solve simultaneously a set of decision problems. At first, the layout of parallel lines must be chosen, deciding if they will be identical and/or independent or not. Second, the products to be manufactured on each line should be determined. Finally each line should be balanced. Solving all these optimization problems jointly results in an extremely complex research challenge.

Another factor influencing the complexity of the problem is the attributes of the tasks that need to be considered. This will be discussed in the next section.

4. Tasks and their attributes

Because the cycle time constraint is commonly used in line balancing, an important characteristic of a task is its processing time. Nevertheless, tasks may have a number of other attributes that must be taken into account when assigning them to workstations such as process and ergonomic parameters, probability of failure, cost, etc. Such attributes may have constant/uncertain/dynamic values or be functions of other decisions made.

Note that in some publications on disassembly line balancing, authors use the term "parts" instead of tasks. This is because parts are removed from end-of-life products. In the present survey, "task" will be used independently of industrial context; for the mentioned case in disassembly, such a task corresponds to the removal of the corresponding part.

4.1. Task attributes with constant values

The value of task attribute j for each task i is given, unique and invariable, $t_j(i)=\text{const}$. The values of t_j may belong to: (i) the set of real or natural numbers, $t_j(i)\in R_+$ or $t_j(i)\in N$, or (ii) a limited set of elements, $t_j(i)\in A$. Table 1 presents the examples of such task attributes that can be found in the literature.

4.2. Task attributes with dynamic values

Such a task attribute is given by its initial value and a function employed to calculate its value at moment t, i.e., $t_j(i)=f(t)$. Such functions (Biskup, 1999; Womer, 1979; Yelle, 1979) are used especially for the formalization of learning and/or linear deterioration effects observed for workers (Boucher, 1987; Chakravarty, 1988; Cohen and Dar-El, 1998; Digiesi et al., 2009; Toksarı et al., 2008, 2010).

4.3. Task attributes with uncertain values

Often the values of task attributes are not known exactly at the point when line balancing decisions have to be made. For example, in manual lines, the effectiveness of operators vary with work rate, skill level, and motivation, which may affect processing times. Numerous other factors such as possible changes in product and workstation characteristics increase the uncertainty of task attributes.

There exist several ways to model uncertain task attributes:

– Random variables with known distributions of probabilities and their parameters, e.g., mean value μ and standard deviation σ,

Table 1
Task attributes with constant values.

Task attribute	References
Examples for $t_j(i) \in R_+$ or $t_j(i) \in N$	
Time	Altekin and Kandiller (2008), Salveson (1955), Sarker and Shanthikumar (1983), Wilhelm (1999)
Processing cost	Altekin and Kandiller (2008), Amen (2000, 2006), Bukchin and Rabinowitch (2006), Hamta et al. (2011), Kazemi et al. (2011)
Incompletion cost	Gamberini et al. (2006, 2009), Kottas and Lau (1973, 1981), Sarin et al. (1999)
Duplication cost	Kara et al. (2011a)
Space	Altemeier et al. (2010), Bautista and Pereira (2011), Chica et al. (2010a, 2010b, 2011a, 2011b, 2011c, 2012, Grangeon et al. (2011)
Number of disassembled parts	Altekin and Kandiller (2008)
Demand for disassembled parts	McGovern and Gupta (2006, 2007a, 2007b, 2011)
Probability of task failure	Gungor and Gupta (2001)
Number of processing alternatives	Bock (2008)
Ergonomic parameters	Choi (2009), Gunther et al. (1983), Otto and Scholl (2011)
Examples for $t_j(i) \in A_+$ or $t_j(i) \in N$	
Execution side of the line or workpiece	Altemeier et al. (2010), Baykasoğlu and Dereli (2008), Hu et al. (2008, 2010), Kim et al. (2009), Tapkan et al. (2011)
Workpiece orientation	Battaïa et al. (2012a), Essafi et al. (2010a, 2010b, 2012)
Required conveyor height	Grangeon et al. (2011), Lapierre and Ruiz (2004)
Equipment/tool used	Ege et al. (2009), Tseng et al. (2008)
Task execution direction	McGovern and Gupta (2006, 2007a, 2007b, 2011)
Property of being hazardous	McGovern and Gupta(2006, 2007a, 2007b, 2011)
Relevant product option	Bock (2008)

i.e., $t_j(i) = f(\mu, \sigma)$, This is often used to model task times in the manual assembly environment (Ağpak and Gökçen, 2007; Bagher et al., 2011; Baykasoğlu and Ozbakır, 2007; Cakir et al., 2011; Dolgui and Proth, 2010; Fazlollahtabar et al., 2011; Gamberini et al., 2009; Özcan, 2010; Özcan et al., 2011).
– Fuzzy numbers with known membership functions, i.e., $t_j(i) = \tilde{\boldsymbol{p}}$. This was used for the task times in Hop (2006), Tsujimura et al. (1995) and Zacharia and Nearchou (2012).
– Scenario definition: if the set of potentially achievable scenarios over a pre-specified planning horizon can be identified, then the values of task attributes can be defined for each possible scenario, i.e., $t_j(i) = \{S\}$, (e.g., for task times, Dolgui and Kovalev, 2012, and Xu and Xiao, 2009, 2011).
– Definition by intervals: the left- and right-hand sides of such intervals determine, respectively the minimal and maximal possible values of the corresponding task attribute, i.e., $t_j(i) = [a,b]$. Its definite value can be known only at the moment of line exploitation. Such task processing times were considered by Gurevsky et al. (2012b).

The choice of one of these models depends on available input data and the line balancing objectives.

4.4. Task attributes with assignment-dependent values

A set of possible values is known for attribute j of task i. Its actual value depends on the values of the decision variables in the current partial solution, such as:

- The set of tasks assigned to a previous or current workstation, i.e., $t_j(i) = f(I_K)$. Here are several examples:
  – Time of task i depends on the set of tasks that are executed previously (Capacho and Pastor, 2006; Capacho et al., 2009; Chen et al., 2002; Kulak et al., 2008; Scholl et al., 2008);
  – Probability of not completing task i depends on the set of tasks executed before at the same workstation (Gamberini et al., 2006, 2009);
  – Time and cost of task i depend on its follower task g (Graves and Lamar, 1983; Wilhelm, 1999).
  – The actual task time and cost depend implicitly on task assignment. For example, machining parameters (and so task times) may depend on the set of tasks performed simultaneously with the same equipment (Battaïa et al., 2012b; Dolgui et al., 2006a, 2006b, 2006c, 2008c; Guschinskaya et al., 2008, 2009).
  – A task time and cost can be related by a function. The shorter the task time is, the greater the task cost (Hamta et al., 2011). The sum of task times assigned to each workstation and total task cost must be minimized concurrently. In the case of a conflict between these goal functions, a compromise can be found.
- The attributes of workstation k where task i is assigned, i.e., $t_j(i) = f(W(k))$, e.g., task time (Chiang et al., 2007; Gao et al., 2009; Kara et al., 2011b; Pekin and Azizoglu, 2008; Yoosefelahi et al., 2012) and/or task cost (Graves and Redfield, 1988; Wei et al., 1997; Yoosefelahi et al., 2012) may depend on the type of equipment installed at workstation k, skill of worker assigned to this workstation (Blum and Miralles, 2011; Wong et al., 2006), or crew size allocated (Kara et al., 2011b; Shtub, 1984; Wilson, 1986).
- The complementary decision variables $V(i)$ used to express decisions related to the tasks, i.e., $t_j(i) = f(V(i))$. A number of task attributes that depend on the task processing alternative were introduced in Bock (2008), such as the total investments necessary for task i, variable payments per period for processing task i, minimum number of workers required for processing task i. Processing alternatives in terms of resources were considered by Corominas et al. (2011).
- The line to which a task is assigned in the case of multiple lines, i.e., $t_j(i) = f(n)$. For example, processing time, task distance and travel time were addressed by Sparling (1998).

The above listed situations are rarely combined in the literature. Only a few examples exist. In Bock (2008), a task time depends on the alternative selected for the task processing and on the number of workers assigned to the workstation. In Wilhelm (1999), task times and costs depend on the tasks that follow and on the workstation selected.

Table 2 presents the publications with models containing more than one task attribute. The numbers in this table represent the number of task attributes of each type considered.

Besides the task attributes, other input data can be associated with tasks. Some kinds of relations between task i and other tasks $I\backslash\{i\}$ can be expressed by a set of tasks participating in this

Table 2
Publications with models containing more than one task attribute.

References	R_+ or N	A	$f(\mu,\sigma)$	$f(I_k)$	$f(W(k))$	$f(V(i))$	$f(W(k),I_k)$	$f(W(k),V(i))$	$f(n)$
Bautista and Pereira (2007, 2011), Chica et al. (2010a, 2010b, 2011a, 2011b, 2011c, 2012), Kara et al. (2011a), Kazemi et al. (2011)	2								
Altekin and Kandiller (2008)	3								
Otto and Scholl (2011)	15								
Choi (2009)	27								
Baykasoğlu and Dereli (2008), Grangeon et al. (2011), Hu et al. (2010), Özcan and Toklu (2009b), Wu et al. (2008)	1	1							
Johnson (1983), Lapierre and Ruiz (2004), Lapierre et al. (2006), Tseng et al. (2008)	1	2							
Gamberini et al. (2009)	2		1						
McGovern and Gupta (2006, 2007a, 2007b, 2011)	2	2							
Gamberini et al. (2006)	3		1						
Altemeier et al. (2010)	3	1			1				
Essafi et al. (2010a, 2010b, 2012)		1		1					
Battaïa et al. (2012a)		1		2					
Gadidov and Wilhelm (2000)		1			2				
Pinnoi and Wilhelm (1997b)		2			2				
Battaïa et al. (2012b), Dolgui et al. (2006a, 2006b, 2006c, 2008c), Guschinskaya et al. (2008, 2009), Hamta et al. (2011)				2					
Graves and Redfield (1988), Kara et al. (2011b), Yoosefelahi et al., 2012, Zhang and Gen (2011)					2				
Bock (2008)						4		1	
Wilhelm (1999)							2		
Sparling (1998)									3

relation, e.g., direct followers of i, tasks incompatible with i (Essafi et al., 2010c; Guschinskaya and Dolgui, 2009), complementary to i (Pinnoi and Wilhelm, 1997b), synchronous with i (Özcan and Toklu, 2009b, 2009c; Rabbani et al., 2012; Simaria and Vilarinho, 2009), tasks disabled by the failure of i (Gungor and Gupta, 2001). Such sets can also depend on the current partial solution. For example in Bock (2008), the set of followers depends on the processing alternatives chosen.

Finally, it should be noted that the possibility to incorporate additional attributes of tasks to existing models if necessary is an important issue. Unfortunately, only a few authors (Boysen and Fliedner, 2007; Johnson, 1983; Kim et al., 2000; Sphicas and Silverman, 1976) have addressed this concern, even if some techniques employed are appropriate to do so, e.g., those based on searching the shortest path in a specific graph.

5. Workstations and their attributes

In the SALBP, only tasks are assigned to workstations. Equipment and other resources are selected at subsequent decision stages. Obviously, solving these problems jointly should improve the final solution. However, this will inevitably increase problem complexity.

If all the tasks assigned to a workstation are related to all the resources allocated to it, then such resource allocation can be characterized by a workstation scalar-attribute $W^s(k)$. It is generally defined on the set of natural numbers, on a limited set, or can take Boolean values. Workstation scalar-attributes have been used in the literature to represent:

- The number of identical parallel workstations (machines or workers) employed (Bock, 2008; Dou et al., 2011; Essafi et al., 2010a, 2010b, 2010c, 2012; Pinto et al., 1981; Yazgan et al., 2011).
- A piece of equipment (Bukchin and Tzur, 2000; Essafi et al., 2010a; Gadidov and Wilhelm, 2000; Yoosefelahi et al., 2011) or a worker selected from a group (Blum and Miralles, 2011; Miralles et al., 2008).
- Buffer capacity associated with the corresponding workstation (Battini et al., 2008).

On other hand, if not all resources assigned to the workstation are used by every task, then a workstation vector-attribute $W^v(k)$ is introduced. It defines which resources are used by which tasks. The number of elements in $W^v(k)$, i.e., dim $W^v(k)$ shows how many resources are assigned to the workstation. A subset of tasks associated with each element defines the tasks that require the corresponding resource. Of course, several vector-attributes can be used if necessary (for example where there are different types of independent resources: machines, workers, etc).

The task partition in subsets can be:

- Straightforward, if there is only one obvious way to partition the tasks assigned to the same workstation into subsets associated with the workstation resources. For example, if each task requires specific equipment (Kim and Park, 1995) or a simple rule can be used to select a resource for each task (e.g., the one with the minimal processing cost or time, see (Chiang et al., 2007; Ege et al., 2009; Gao et al., 2009), then the decision about what resources are used by which tasks is straightforward. Another example is the case of lines with multiple parallel workplaces. If each task is associated with a unique side of the workpiece, then all tasks associated with the same side must be assigned to the same workplace. Thus, the partition of the set of workstation tasks in subsets is evident (Battaïa et al., 2012a, 2012b; Kahan et al., 2009; Scholl and Boysen, 2009).
- A combinatorial optimization problem, if there exists a number of possible partitions, e.g., for two-sided lines where some tasks can be executed both on the left and right sides of the line (Kim et al., 2009; Özbakır and Tapkan, 2011; Özcan, 2010; Özcan and Toklu, 2009b, 2009c; Tapkan et al., 2011). In this case, some additional constraints are imposed, like inclusion or exclusion relations (Belmokhtar et al., 2006, 2007; Borisovsky et al., 2012a, 2012b; Delorme et al., 2012). The criteria to be optimized, usually consider the total number of resources employed, their total cost or number of workstations (Chang and Chang, 2010; Dimitriadis, 2006; Fattahi et al., 2011; Hu et al., 2008; Moon et al., 2009; Simaria and Vilarinho, 2009). Theoretically, each subset of a workstation vector-attribute may be partitioned into its own subsets and so on, like in the

lines with series-parallel activation of workplaces studied in Dolgui and Ihnatsenka (2009a).

6. Problem constraints

Constraints are used to distinguish feasible assignments of tasks to workstations. They can have technological, economic or ergonomic origins. They can also express some preferences of the decision maker. In this case they are not exactly constraints but the system designer has decided to add them by experience. In mathematical models, the constraints must be expressed explicitly in linear or non-linear form. In a solution procedure, they can be taken into account either during the solution's construction or for a solution obtained. In this section, the principal constraints used in flow line balancing problems are highlighted.

6.1. Assignment constraints

The following constraints are used to express some relations among decision variables.

Occurrence constraints: All tasks from set I must be executed either in a cycle as for single-product lines, or in several cycles as in multi- or mixed- product lines (Bock, 2008; Hwang and Katayama, 2009; Kara et al., 2007; Venkatesh, 2008; Xu and Xiao, 2009; Zhang and Gen, 2011).

Precedence constraints are given by a simple precedence graph or a precedence matrix with 0–1 values (Salveson, 1955). They depend on which process plans were selected before. Usually process planning is executed separately, prior to line balancing. Integrating process planning and line balancing can increase the efficiency of the solutions obtained. Diverse models for precedence constraints have been presented in literature to incorporate process plan selection into line balancing problems. Gungor and Gupta (2001) as well as Altekin and Kandiller (2008) have used an AND/OR precedence graph expressing more sophisticated precedence requirements like OR, complex AND/OR and XOR relationships. Capacho and Pastor (2006) as well as Capacho et al. (2009) have introduced a hypergraph having alternative precedence subgraphs. If-then rules have been employed by Topaloglu et al. (2012). Notice that how the precedence constraints are considered is often the only difference between straight and U-line balancing problems, since for the latter, the worker can perform tasks assigned to different sides of the line (Ağpak and Gökçen, 2007; Hwang and Katayama, 2009; Hwang et al., 2008; Kara et al., 2007; Sabuncuoglu et al., 2009; Toklu and Özcan, 2008; Toksarı et al., 2008).

Inclusion/Conjunctive/Positive Zoning constraints are used to force the assignment of a set of tasks to the same workstation. Only if no workstation vector-attributes exist and values of task attributes for different tasks can be cumulated, then such a set of tasks can be replaced with, macro-task (Buxey, 1974; Deckro, 1989), otherwise this is impossible (Akpınar and Bayhan, 2011; Baykasoğlu and Dereli, 2008; Dolgui et al., 2005; Guschinskaya and Dolgui, 2009; Guschinskaya et al., 2008; Özbakır and Tapkan, 2011; Özcan, 2010; Özcan and Toklu, 2009c; Purnomo et al., 2013; Simaria and Vilarinho, 2009; Tapkan et al., 2011).

Exclusion/Disjunctive/Negative Zoning constraints are employed to forbid an assignment of certain tasks to the same workstation (Baykasoğlu and Dereli, 2008; Dolgui et al., 2005; Özbakır and Tapkan, 2011; Özcan, 2010; Purnomo et al., 2013; Simaria and Vilarinho, 2009) or to the same element of a workstation vector-attribute (Akpınar and Bayhan, 2011; Dolgui et al., 2005; Dolgui et al., 2008c; Guschinskaya and Dolgui, 2009; Guschinskaya et al., 2008; Li et al., 2011; Tapkan et al., 2011).

Synchronism constraints schedule the execution of certain related tasks on parallel workplaces (Özcan and Toklu, 2009b, 2009c; Purnomo et al., 2013; Rabbani et al., 2012; Simaria and Vilarinho, 2009; Tapkan et al., 2011).

Distance constraints impose a distance between tasks (minimum or maximum). Such distances may be expressed in terms of the number of workstations or a difference between their starting times (Park et al., 1996; Pastor and Corominas, 2000; Purnomo et al., 2013).

Positional constraints prohibit the assignment of some tasks to one or several specific workstations (Kim et al., 2000; Purnomo et al., 2013; Tapkan et al., 2011).

6.2. Constraints concerning workstation attributes

Constraints between workstation and task attributes express the dependencies between the value of a workstation attribute j and the tasks assigned to workstation k, i.e., $W_j(k) = f[I(k)]$.

- If a workstation scalar-attribute $w_j^s(k)$ depends directly on a task attribute t_h, then the first task assigned to the workstation determines the value of this workstation attribute. It will be equal to the value of the corresponding task attribute (the values of both attributes must belong to the same limited set A). For example, if only one workpiece side is accessible at each workstation, then this side must correspond to the side required for all tasks assigned to it (Grangeon et al., 2011; Lapierre and Ruiz, 2004).
- The value of a workstation scalar-attribute can be calculated on the basis of the attributes of tasks assigned to the workstation. For example, to define the number of workers allocated to a workstation k, i.e., w_k, the following rule can be used: $w_k = [T_k/c]$ (McMullen and Frazier, 1998; McMullen and Tarasewich, 2003, 2006).

Constraints among workstation attributes are employed to prevent infeasible combinations of workstation attribute values for the same or different workstations. For example, such constraints have been used to forbid the allocation of incompatible equipment to the same workstation (Borisovsky et al., 2012a, 2012b; Corominas et al., 2011; Delorme et al., 2012; Dolgui et al., 2006c; Rekiek and Delchambre, 2006). On the other hand, when there exists a set of available equipment (Belmokhtar et al., 2006; Borisovsky et al., 2012a, 2012b; Delorme et al., 2012; Kara et al., 2011b) or a crew of workers with diverse skill levels (Blum and Miralles, 2011; Chan et al., 1998; Kara et al., 2011b; Miralles et al., 2008), and if one of these unique resources is allocated to a workstation, then it becomes unavailable for other workstations. Another factor in such constraints is the existence of working zones characterized by the same value of certain workstation scalar-attributes like the conveyor height considered in Lapierre and Ruiz (2004). The values of scalar-attributes of different workstations can be independent as well. For example, if it is supposed that the equipment to be installed at workstations has to be purchased, then, the number of pieces of equipment is not limited. Thus, the same equipment type can be used at all workstations if necessary (Bukchin and Rubinovitz, 2002; Bukchin and Tzur, 2000; Gadidov and Wilhelm, 2000; Graves and Redfield, 1988).

6.3. Constraints on performance measures calculated for workstations

The following constraints may exist on the performance measures $w_j(k)$ calculated on the basis of the values of workstation and task attributes:

- $p_j(k) \le b$, where b is a constant, e.g., cycle time constraint, or limitation on the space required for executing the tasks assigned to the workstation (Sawik, 2002), or ergonomic constraints considered by Otto and Scholl (2011).

- $p_j(k) \le b$, where b is a constant, e.g., the probability of on-time completion of all tasks assigned to the workstation which must not be inferior to a given value (Ağpak and Gökçen, 2007; Baykasoğlu and Ozbakır, 2007; Dolgui and Proth, 2010; Urban and Chiang, 2006).
- $p_j(k) \le w_j^s(k) \cdot b$ where b is a constant. Such constraints were used to express the cycle time constraint for the case of identical parallel equipment (Bukchin and Rubinovitz, 2002; Essafi et al., 2010b, 2010c; McMullen and Frazier, 1998) or for the case where an unbalance among workstations is desired (Erel and Gokcen, 1999; Johnson, 1983).

6.4. Cycle time constraint

Workstation processing times have often to be less than an objective value calculated on the basis of the required line throughput. This is known as the cycle time constraint. Nevertheless, the formula to calculate workstation processing times is not always simple. This depends on the line layout, equipment used, etc. If multiple workplaces are employed, then the processing time depends on how they are activated: sequentially, in parallel or in a mixed series-parallel manner (see Fig. 3).

Sequentially executed tasks. The following situations have been reported in the literature.

- Processing time of a workstation is equal to the sum of the processing times of all their tasks. Sometimes, a supplementary time (increment) representing workpiece loading/unloading is also added (Boysen et al., 2007).
- Processing time of a workstation is equal to the sum of the processing times of all tasks assigned to this workstation plus sequence-dependent auxiliary times for the change or displacement of tools, workpiece rotations, etc. (Andrés et al., 2008; Essafi et al., 2010b, 2010c; Martino and Pastor, 2010; Nazarian et al., 2010; Seyed-Alagheband et al., 2011). Such auxiliary times can be introduced in a matrix form as estimated setup times occurring between each pair of tasks. If a sequence of two tasks is forbidden by the precedence constraints, then the auxiliary time between them are considered as infinite (Arcus, 1966; Essafi et al., 2010b, 2010c). In Graves and Redfield (1988) and Wilhelm (1999), a tool is associated with each task and an auxiliary time is added when a tool replacement is needed. Nazarian et al. (2010) have pointed out that the inter-task times between different product models (inter-model inter-task times) are often larger than inter-task times for the same model (intra-model inter-task times).

Simultaneously executed tasks. There are a number of parallel workplaces at each workstation and tasks are executed simultaneously at all workplaces, i.e., all tasks assigned to a workstation are processed at the same time and, as a consequence, workstation processing time is equal to the longest processing time for the tasks assigned. Such a situation is possible for the machining lines with parallel multi-spindle heads as has been studied in Belmokhtar et al. (2006), Borisovsky et al. 2012a, 2012b, Delorme et al. (2012), Dolgui et al. (2006c, 2008b, 2009a, 2009b).

Mixed mode for execution of tasks. Even if any combination of previously presented cases is theoretically possible, as of yet only the following ones have been considered in the literature.

- There are a number of parallel workplaces and the tasks assigned to each workplace are executed sequentially (Baykasoğlu and Dereli, 2008; Becker and Scholl, 2009; Chang and Chang, 2010; Hu et al., 2010; Kahan et al., 2009; Kim et al., 2000, 2009; Lee et al., 2001; Özcan and Toklu, 2009b, 2009c, 2009d; Simaria and Vilarinho, 2009; Wu et al., 2008). In this case, a workstation processing time is equal to the longest time among processing times of its workplaces. A workplace processing time is equal to the sum of processing times of its tasks plus slack times that can occur due to precedence relations between some operations assigned to the parallel workplaces (Fattahi et al., 2011; Lee et al., 2001). The sequence-dependent auxiliary times have also been considered for this case by Özcan and Toklu (2010).
- There are a number of serial workplaces, but the tasks assigned to each workplace are executed simultaneously (Dolgui et al., 2006a, 2005; Dolgui and Ihnatsenka, 2009b; Guschinskaya and Dolgui, 2009; Guschinskaya et al., 2008). In this case, a workstation processing time is equal to the sum of the processing times for its workplaces. Processing time of a workplace is equal to the longest time of its tasks (Guschinskaya and Dolgui, 2009) or calculated in a more sophisticated manner taking into account the parallel execution of tasks with interdependent task characteristics (Dolgui et al., 2009b).
- Workplaces are activated in a serial-parallel way and the tasks assigned to each workplace are performed simultaneously (Belmokhtar et al., 2007; Dolgui and Ihnatsenka, 2009a; Guschinskaya et al., 2009).

7. Objective functions

An objective or goal function evaluates the quality of feasible solutions and is used to choose the best one. The following three goal functions are widely used for SALBPs.

- Minimize the number of workstations m to be installed (SALBP-1). This objective is the most frequently employed in literature, e.g., (Bautista and Pereira, 2009; Chiang et al., 2007; Kilincci, 2011).
- Minimize line cycle time c (SALBP-2), e.g., (Gao et al., 2009; Kilincci, 2010; Kim et al., 2009; Kulak et al., 2008; Miralles et al., 2008; Nearchou, 2007; Seyed-Alagheband et al., 2011).
- Maximize line efficiency (E), i.e., the value $m \cdot c$ (SALBP-E). This objective is less used and more difficult to deal with because of its nonlinear form (Chakravarty, 1988; Macaskill, 1972; Scholl and Klein, 1999; Wei and Chao, 2011). For mixed-model lines, a weighted efficiency of the line can be optimized taking into account the ratios of models being produced (Vilarinho and Simaria, 2006).

The impact of selecting one of the above objective functions was considered in Bukchin (1998) and Nkasu and Leung (1995).

Among others, the following goal functions are also frequently used:

- Maximize the "system utilization" U which is another measure of "line efficiency" and can be calculated as the ratio of the minimum required number of workers to those actually assigned (McMullen and Frazier, 1997; McMullen and Tarasewich, 2003; Vilarinho and Simaria, 2006).
- Minimize *line smoothness index* (SI). None of previous functions warrant a solution with well-balanced workstations, i.e., having similar workstation processing times. The assessment of the quality of a balance can be based on:
  - (i) absolute/squared/average deviations: $\sum_{k=1}^{m} |T_k - \tilde{T}|$, $\sqrt{\sum_{k=1}^{m} |T_k - \tilde{T}|^2}$, $\sum_{k=1}^{m} |T_k - \tilde{T}|^2$;

Table 3
Objective functions considered jointly.

References	M	$\sum \dim W^v(k)$	c	$\lvert I_k \rvert$	SI	E	IWR	Cum	Re	Pr
Özcan and Toklu (2009c, 2010)	1	2								
Chutima and Chimklai (2012)	1	2			3		3			
Kara et al. (2011a), Özcan and Toklu (2009d)	1		2	3						
Otto and Scholl (2011), Simaria and Vilarinho (2009), Vilarinho and Simaria (2006), Yu and Yin (2010)	1				2					
McGovern and Gupta (2006, 2007a, 2007b, 2011)	1				2			3,4,5		
Özbakır and Tapkan (2011)	2				1					
Fattahi et al. (2011)	2							1		
Kovalev et al. (2012)	1								2	
Baykasoglu (2006)					1	2				
Purnomo et al. (2013)					2	1				
Toklu and Özcan (2008)	+		+	+						
Brudaru et al. (2010)	+				+					
Yagmahan (2011)	+				+,+					
Chutima and Olanviwatchai (2010)	+				+		+			
Ding et al. (2010)	+				+			+		
Grangeon et al. (2011), Yang et al. (2011)	+				+				+	
Bagher et al. (2011)	+				+					+
Chica et al. (2010a, 2010b, 2011a, 2011b, 2011c), Pekin and Azizoglu (2008)	+							+		
McMullen and Tarasewich (2006)		+				+		+		+
Nearchou (2008, 2011), Nourmohammadi and Zandieh (2011), Zacharia and Nearchou (2012)			+		+					
Hamta et al. (2011)			+					+		
Yoosefelahi et al. (2012)			+					+,+		
Zhang and Gen (2011)			+		+			+		
Hwang and Katayama (2009), Hwang et al. (2008), Özcan and Toklu (2009b)					+	+				
Cakir et al. (2011)					+			+		
Tseng et al. (2008)					+			+,+		
Gamberini et al. (2006, 2009)								+	+	
McMullen and Frazier (1998)								+		+

– (ii) comparison with the maximum/average workstation processing time ($\tilde{T}$) or with the desired cycle time ($\tilde{T}=c$ or $\tilde{T}=1/m\sum_{i\in I}t_i$);
– (iii) deviations for all workstations for the same product (Eswaramoorthi et al., 2012; Mozdgir et al., 2013), for the same workstation but for different products (Kara et al., 2007) or both of them (Yang et al., 2011).

- Options (i)–(iii) may give numerous criteria. A weighted sum of them can be calculated to obtain one sole criterion. Several computational evaluations of objectives to smoothen workload have been presented (Emde et al., 2010; Venkatesh, 2008). Additionally, ''smoothness'' may concern other performance measures, not necessarily only processing times, e.g., workstation ergonomic risks (Otto and Scholl, 2011).
- Minimize $\sum_{k=1}^{m} dimW^v(k)$ if workstation vectorial attributes are employed (Simaria et al., 2009, Vilarinho and Simaria, 2006).
- Maximize the probability of on-time completion of all tasks i.e., $Pr=\prod_{k=1}^{m} Pr[p_j(k)\le c]$, where $Pr[p_j(k)\le c]$ represents the probability of on-time completion of stochastic tasks assigned to workstation k (Baykasoğlu and Ozbakır, 2007; Dolgui and Proth, 2010; McMullen and Tarasewich, 2006). This objective can be also expressed as a minimization of non-completion probabilities of each station (Bagher et al., 2011).
- Maximize the cohesion of the tasks (IWR) assigned to the same workstations. This goal is used if the tasks related to the same product's function should be assigned to the same workstation (Agrawal, 1985). An index of work relatedness (Agrawal, 1985; Genikomsakis and Tourassis, 2010) and a task proximity index (Genikomsakis and Tourassis, 2012) have been proposed to measure this cohesion.
- Minimize or maximize $Cum=\sum_{k=1}^{m} f_1(T(k))+\sum_{k=1}^{m} f_2(W(k))$, calculated on the basis of task ($T(k)$) and workstation ($W(k)$) attributes, where Cum stands for cumulative objective function. Functions f_1 and f_2 are not always linear. They can be cost-oriented to be minimized or benefit-oriented to be maximized, for example: cost of an assembly line (Kara et al., 2011b); cost of a machining system (Battaïa et al., 2012a, 2012b; Borisovsky et al., 2012a, 2012b; Dolgui et al., 2012; Essafi et al., 2010a, 2012, 2010c; Guschinskaya et al., 2008, 2009, 2011); cost of a reconfigurable manufacturing system (RMS) (Delorme et al., 2009; Dou et al., 2011); profit (Altekin and Akkan, 2012); area occupied by workstations (Chica et al., 2010a, 2010b, 2011a, 2011b, 2011c, 2012); hazard, demand and direction measures (McGovern and Gupta, 2006, 2007a, 2007b, 2011); and ergonomic risks (Otto and Scholl, 2011).
- Minimize the reconfiguration cost, denoted by Re, which can be expressed by setup costs (Kovalev et al., 2012), difference between initial and new solutions (Gamberini et al., 2006, 2009), number of transferring operations (Grangeon et al., 2011) or cost for the retraining of operators (Yang et al., 2011).

Often, a line balancing problem can have several objectives in conflict. In this case multi-objective techniques should be used. Table 3 shows what are the combinations of objective functions considered in the multi-objective line balancing problems studied in literature. The numbers give the order of priority for the objectives considered. The character "+" is used if all objectives have the same priority. The methods proposed for multi-objective decision making will be presented in Section 9.

8. Solution methods

Solution methods are often divided into two categories: exact or approximate. Since even the core line balancing problem, SALBP, is NP-hard, the required computational time for obtaining an optimal solution with an exact method for most of line balancing problems increases exponentially with the size of instance considered. As a consequence, approximate methods are clearly needed in order to cope with large scale cases. Additionally, simulation can help to evaluate the dynamic behaviour of the line (McMullen and Tarasewich, 2003; Mendes et al., 2005; Sheu and Chen, 2008; Cortés et al., 2010).

8.1. Exact methods

Line balancing problems can be solved optimally via one of the two following approaches: using a standard general solver (like ILOG Cplex, ILOG Solver, Dash Xpress MP, LINGO, OSL, etc.) or an original dedicated solution method. In the first case, the goal is to define an appropriate mathematical model for the problem and to adjust the solver parameters in order to solve this as quickly as possible. The following mathematical models have been described in literature:

- Mixed integer programs, usually solved with the well-known Branch and Bound or Branch and Cut methods. The following solvers have been used in computational tests: ILOG Cplex (Altekin and Kandiller, 2008; Andrés et al., 2008; Battaïa and Dolgui, 2012; Choi, 2009; Corominas et al., 2008; Delorme et al., 2012; Kim et al., 2009; Miralles et al., 2007; Pastor, 2011; Seyed-Alagheband et al., 2011), LINGO (Nazarian et al., 2010; Paksoy et al., 2012; Toksarı et al., 2010), Xpress MP (Scholl and Boysen, 2009), GAMS (Borisovsky et al., 2012b; Özcan and Toklu, 2010), OSL (Gadidov and Wilhelm, 2000; Pinnoi and Wilhelm, 1997a). The column generation method has been applied with OSL (Wilhelm, 1999). Linear programming (Altekin and Kandiller, 2008; Pastor and Ferrer, 2009) and Lagrangian relaxations (Aghezzaf and Artiba, 1995) have also been used.
- A nonlinear integer program was formulated by Hamta et al. (2011) and solved with LINGO's global solver.
- Goal and fuzzy goal programs were studied using GAMS in Özcan and Toklu (2009d).
- Chance-constrained programs, results obtained with GAMS were reported in Ağpak and Gökçen (2007), Özcan (2010), Urban and Chiang (2006).
- Constraints satisfaction programs, figures obtained with ILOG Solver can be found in Topaloglu et al. (2012).

Since solvers are designed to cope with a large class of optimization problems, they may not be sufficiently efficient when dealing with certain versions of the line balancing problems or even for a particular structure of input data. In this case, an original dedicated exact method can be developed such as:

- Branch and Bound, besides the most referenced algorithms for SALBP such as FABLE (Johnson, 1988), EUREKA (Hoffmann, 1992), SALOME (Scholl and Klein, 1997), a number of new methods are actually available for different line balancing problems (Chiang et al., 2007; Dolgui and Ihnatsenka, 2009a, 2009b; Ege et al., 2009; Hu et al., 2010; Liu et al., 2008; Miralles et al., 2008; Pekin and Azizoglu, 2008; Sewell and Jacobson, 2012; Wu et al., 2008);
- Dynamic programming, see for example Bard (1989), Held et al. (1963), Jackson (1956), is often based on the constrained shortest path in a specific graph with partial solutions (Dolgui et al., 2006c, 2008c; Gungor and Gupta, 2001; Gutjahr and Nemhauser, 1964).

Often, different exact methods are developed for the same problem, since each may be especially efficient for particular structures of input data. The effectiveness of an exact method is usually measured by the computation time required to solve a problem. In Guschinskaya and Dolgui (2009), the authors conclude that if the density of constraints is rather high then the shortest path method is more rapid whereas integer linear programming with ILOG Cplex is quicker if this density is low.

Since the line balancing problem is NP-hard, the utilization of approximate methods is inevitable for large-scale instances or if the available time is strongly limited by decision contexts.

8.2. Approximate methods

Approximate methods do not guarantee optimality, but usually are able to achieve good feasible results in an acceptable computation time. Such approaches can be divided into: bounded exact methods, simple heuristics, and metaheuristics.

8.2.1. Bounded exact methods

Such approximation methods perform an incomplete enumeration of the solution space. They can be obtained by bounding existing exact methods either by restricting the explored solutions space or by limiting the available resolution time. In Bautista and Pereira (2009, 2011) and Ege et al. (2009), the graph of solutions has been truncated. In Bukchin and Tzur (2000) and Miralles et al. (2008), not all vertices are explored by the branch and bound algorithm proposed. In Blum and Miralles (2011), a beam search technique has been applied. A similar approach was employed using Petri nets (Kilincci, 2010, 2011; Kilincci and Bayhan, 2006, 2008), and in the heuristic proposed by Hoffmann for SALBP (Hoffmann, 1963), as well as in its modifications (Fleszar and Hindi, 2003) and adaptations for other types of line balancing problems (Dimitriadis, 2006; Hu et al., 2008).

8.2.2. Simple heuristics

They are usually confined to a particular problem. In the majority of cases, priority rules are used to assign tasks. The most employed are based on task attributes, such as task time or number of followers. Such approaches can be found in Capacho et al. (2009), Pastor et al. (2012), Scholl and Becker (2006), for example. Many of these rules were also adapted for more complex problems to take into account a larger number of task attributes and workstation characteristics. Composite priority rules, where several rules are considered in lexicographic order, have also been proposed in Boctor (1995), Raouf and Tsui (1982). The category of simple heuristic methods can be divided into the following two classes:

- Single-pass heuristics. Only one iteration is made to assign tasks using a greedy function or a priority rule (Raouf and Tsui, 1982; Toksarı et al., 2008). The solution is the final output. As a consequence, solution time is very short even for large-scale problems.
- Multi-pass heuristics. Due to the inherent randomness in these algorithms, differing results can be obtained at each pass and the best solution found after a number of iterations is returned as output (Andrés et al., 2008; Cortés et al., 2010; Finel et al., 2008; Nazarian et al., 2010; Yegul et al., 2010). Randomness is used to select a task to be assigned: select at random a task from a list of candidates (Bock, 2008; Jolai et al., 2009; Toksarı et al., 2010), among tasks having the greatest values of a greedy function (Guschinskaya et al., 2011), or according to a random priority rule (Gamberini et al., 2009). The stop criterion may be expressed with a specified number of iterations or a given number of iterations that do not improve the best

existing solution or/and a pre-specified time to reach a solution.

Simple heuristics can be used to provide an upper bound for an exact method (Battaïa and Dolgui, 2012) or be integrated into metaheuristics for local improvements of intermediate solutions (Essafi et al., 2012).

8.2.3. Metaheuristics, hybrid metaheuristics and matheuristics

These approaches can be employed to solve a large number of optimization problems. To apply a metaheuristic to a particular problem, it is necessary to follow the metaheuristic framework. While such methods are both numerous and varied, they can be divided broadly into the following classes:

- Neighbourhood methods such as Tabu search (Lapierre et al., 2006; Özcan and Toklu, 2009b), Kangaroo method (Minzu and Henrioud, 1998), GRASP (Andrés et al., 2008; Chica et al., 2010a; Essafi et al., 2012; Guschinskaya et al., 2011), Simulated annealing (Cakir et al., 2011; Kara et al., 2007; Otto and Scholl, 2011; Özcan, 2010; Özcan et al., 2010; Özcan and Toklu, 2009c; Seyed-Alagheband et al., 2011), etc.
- Evolutionary approaches, such as differential evolution methods (Mozdgir et al., 2013; Nearchou, 2007, 2008; Nourmohammadi and Zandieh, 2011), imperialist competitive algorithms (Bagher et al., 2011), or genetic algorithms (Akpınar and Bayhan, 2011; Baykasoğlu and Ozbakır, 2007; Dou et al., 2011; Gao et al., 2009; Hamta et al., 2011; Hwang and Katayama, 2009; Hwang et al., 2008; Kazemi et al., 2011; Kim et al., 2009; Kulak et al., 2008; Moon et al., 2009; Özcan et al., 2011; Purnomo et al., 2013; Rabbani et al., 2012; Yu and Yin, 2010; Zacharia and Nearchou, 2012; Zhang and Gen, 2011). Cellular multi-grid genetic algorithms were suggested by Brudaru et al. (2010). Often, the standard scheme of genetic algorithms is completed with a local search. In this case, the corresponding hybrid algorithm is called a memetic algorithm (Gamberini et al., 2009; Guschinskaya et al., 2011). A coincidence memetic algorithm was proposed in Chutima and Olanviwatchai (2010). Another example of hybridization of genetic algorithms consists in such a way that an individual represents a set of rules (heuristics) to be used to construct a feasible solution. The genetic algorithm works with sequences of heuristics to be applied and not with the solutions them-self (Baykasoğlu and Ozbakır, 2007; Guschinskaya et al., 2011). A review of the application of genetic algorithms in assembly line balancing was presented by Tasan and Tunali (2008).
- Swarm intelligence based metaheuristics such as Particle swarm optimization algorithms (Chutima and Chimklai, 2012; Nearchou, 2011), Bees algorithms (Özbakır and Tapkan, 2011; Tapkan et al., 2011), Ant colony optimization (ACO) (Bautista and Pereira, 2007; Baykasoğlu and Dereli, 2008; Chica et al., 2010b; Ding et al., 2010; Fattahi et al., 2011; Ozbakir et al., 2011; Sabuncuoglu et al., 2009; Simaria and Vilarinho, 2009; Simaria et al., 2009; Yagmahan, 2011) and its hybridization with Beam search (Blum, 2008), where artificial ants perform a probabilistic beam search in which extensions of partial solutions are done in an ACO fashion.

The most popular algorithms seem to be genetic algorithms, followed by simulated annealing and ant colony optimization. Metaheuristics are also the most intensively applied methods for solving multi-objective line balancing problems. They will be discussed in the next section. A new and promising tendency is matheuristics (Dolgui et al., 2010). These are based on the interaction of metaheuristics and exact methods. In some elements of metaheuristics, features from mathematical models of the problem considered are employed (Dolgui et al., 2010).

9. Multiple-criteria decision making

Searching a compromise for two or more conflicting objectives is often necessary in line balancing. Consequently, as it can be seen in Table 3, multi-criteria approaches have attracted a lot of attention from researchers. Some of the best known methods for solving multi-objective line balancing problems are enumerated and discussed below.

Aggregation of objectives. All of the objective functions are combined into a single function, for example, by using a weighted sum (Akpınar and Bayhan, 2011; Bagher et al., 2011; Brudaru et al., 2010; Fattahi et al., 2011; Gamberini et al., 2009; Hwang and Katayama, 2009; Hwang et al., 2008; Nearchou, 2008; Özbakır and Tapkan, 2011; Özcan et al., 2010; Özcan and Toklu, 2009c; Purnomo et al., 2013; Simaria and Vilarinho, 2009; Simaria et al., 2009; Tseng et al., 2008; Venkatesh, 2008; Wu et al., 2008; Yu and Yin, 2010; Zacharia and Nearchou, 2012). Another approach based on weighted *Lp* distance (*Lp*-metric) from any solution x to the ideal is considered for example by Hamta et al. (2011) for two objectives and $p=1$. For such methods, the final solution depends on relative values of the weights specified. To compute weights for multi-objective problems three principal methods are discussed in the literature:

- Fixed-weight method uses constant weights which satisfy the relation $\sum_{k=1}^{m} w_i = 1$, where k is the number of objectives, w_i is the weight of objective i and $w_i > 0$, $\forall i = 1,\ldots,k$. However, different combinations of weights may be tested for the same set of objectives (Brudaru et al., 2010; Özcan et al., 2010).
- Random–weight method assigns random values to the weights and the same problem is solved several times in order to obtain more diversified non-dominated solutions (Gamberini et al., 2009; Tseng et al., 2008; Zacharia and Nearchou, 2012).
- Adaptive-weight method readjusts the weights by utilizing some useful information from the solutions already obtained (Hwang et al., 2008; Nearchou, 2008; Zacharia and Nearchou, 2012). This method can provide solutions located on the non-convex part of a Pareto front (Kim and de Weck, 2005).

Lexicographic resolution. An order is selected to apply the decision criteria. The solutions are compared with by using the first criterion. If a tie occurs, then the second criterion is employed, and so on (Lee et al., 2001). Another technique consists in solving the problem taking into account the first objective function. The values obtained are used in the constraints at the next step where the second criterion is optimized (Gungor and Gupta, 2001; Vilarinho and Simaria, 2002), and so on. For example, the Lexicographic Bottleneck Assembly Line Balancing Problem (Pastor, 2011; Pastor et al., 2012) minimizes first the workload of the most heavily loaded workstation, and then the workload of the second most heavily loaded workstation, and so on.

Goal (GP) and fuzzy goal programming (FGP). For GP, the target values of all criteria must be precisely defined by the decision maker(s) (Gökçen and Ağpak, 2006; Özcan and Toklu, 2009d). The original objectives are expressed as a linear function of target values. Two types of auxiliary variables are used to represent an under-achievement of the target value with negative deviation (d^-) and an over-achievement of the target value with positive deviation (d^+). If the target values are imprecise, vague, or uncertain, then they can be considered like fuzzy values. Triangular, trapezoidal, or linear membership functions are used to define them (Kara et al., 2010, 2009;

Özcan and Toklu, 2009d; Toklu and Özcan, 2008). GP and FGP can use either lexicographic or weighted additive models as described above.

Pareto-based ranking. Such methods are used for comparing a set of solutions. Methods that have been used in the line balancing context include:

- PROMETHEE II (Rekiek and Delchambre, 2006) was applied to compare a population of individuals created by a genetic algorithm. The principle consists in comparing a solution with a set of other solutions. The results of the comparisons are then used in order to rank the solutions obtained.
- TOPSIS (Total Order Preference by Similarity to the Ideal Solution) (Gamberini et al., 2006; Nourmohammadi and Zandieh, 2011) compares all solutions with an ideal one.
- Data Envelopment Analysis (DEA) is a linear programming methodology to measure the efficiency of multiple decision-making when the production process presents a structure of multiple inputs and outputs (Jolai et al., 2009).
- The solutions can also be ranked using a generalized Pareto-based scale-independent fitness function proposed in Zhang and Gen (2011). These values result in tournament-like scores calculated for all participants.

Constructing Pareto front. These methods search for a set of non-dominated solutions that represent the Pareto front of the problem. A complete set of Pareto-optimal solutions can be obtained by an exact method. For example, a branch and bound was used in Pekin and Azizoglu (2008) for a bi-criteria assembly line design problem. In this problem the total equipment cost and number of workstations were minimized. In the literature, there are publications where an approximation of Pareto front is constructed by using, for example:

- A heuristic (Gamberini et al., 2009);
- A multiple ant colony system (Chica et al., 2010b, 2011b, 2011c);
- Multi-objective ACO (Ding et al., 2010);
- Simulated annealing using a multinomial probability mass function (Cakir et al., 2011);
- Particle swarm optimization algorithms (Chutima and Chimklai, 2012; Nearchou, 2011);
- An evolutionary algorithm: MODE (multi-objective differential evolution) heuristic (Nearchou, 2008), HEMOAs (hybrid evolutionary multiple-objective algorithms) (Tseng et al., 2008), MOGA (multiple-objective genetic algorithm) (Gamberini et al., 2009; Zhang and Gen, 2011), and NSGA-II (Chehade et al., 2012).

The feasible solutions obtained are then compared in order to eliminate those that are dominated. For this purpose, the following algorithms can be used: compare each objective independently or compare a fitness function obtained by an aggregation of the objectives considered (Tseng et al., 2008). An estimated Pareto-optimal frontier should have a sufficient amount of non-dominated solutions, a short distance to the true-Pareto frontier and a sufficient diversity among the solutions (Zitzler and Thiele, 1999). These metrics can be used to compare Pareto-fronts provided by different methods.

10. Discussion

The following trends in the current line balancing literature merit being mentioned.

First of all, it should be stated that numerous recent publications still deal with the basic versions of assembly line problems: SALBP-1 (Bautista and Pereira, 2009; Blum, 2008; Kilincci, 2011; Kilincci and Bayhan, 2008; Liu et al., 2008; Özcan and Toklu, 2009a; Pastor and Ferrer, 2009; Sewell and Jacobson, 2012) and SALBP-2 (Killinci, 2010; Nearchou, 2007; Pastor and Ferrer, 2009). SALBP-E has been less treated, perhaps because of its non-linear objective function. Nevertheless, since a very effective algorithm has been recently developed for SALBP-1 by Sewell and Jacobson (2012), this may provide a promising perspective for solving SALBP-E efficiently in the near future.

A lot of effort has been made in order to expand SALBP by solving it jointly with other problems such as process planning, equipment selection and/or line design. Making several decisions simultaneously can result in better final line performance and effectiveness, but may increase the complexity of the optimization problem. Recent publications have discussed the following joint or multiple optimization problems, including line balancing:

- Process selection and line balancing (Bock, 2008; Capacho and Pastor, 2006; Capacho et al., 2009; Corominas et al., 2011).
- Line design and balancing (AlGeddawy and ElMaraghy, 2010; Bock, 2008; Li et al., 2011; Nazarian et al., 2010; Simaria et al., 2009).
- Line balancing and scheduling at each workstation (Kara et al., 2007; Özcan and Toklu, 2009b, 2009c; Purnomo et al., 2013; Rabbani et al., 2012; Simaria and Vilarinho, 2009; Tapkan et al., 2011),
- Line design, balancing and scheduling (Battaïa and Dolgui, 2012; Battaïa et al., 2012a, 2012b; Borisovsky et al., 2012a, 2012b; Delorme et al., 2012; Essafi et al., 2010c, 2012; Guschinskaya and Dolgui, 2009).

Furthermore, line balancing problems can be incorporated into the design and optimization of the entire supply chain (Paksoy et al., 2012).

Similarly, many of the previously studied balancing problems have now been considered in a multi-objective context (Gamberini et al., 2009; Hwang and Katayama, 2009; Hwang et al., 2008; Özcan and Toklu, 2009c; Simaria and Vilarinho, 2009; Wu et al., 2008). To date, the presence of multiple objectives has tended to be treated with metaheuristics. Nevertheless, the development of effective exact methods seems to be a promising direction for future research.

In regard to solution methods, the development of new Branch and Bound algorithms for more complex problems (Dolgui and Ihnatsenka, 2009a, 2009b; Miralles et al., 2008; Wu et al., 2008) and frequent use of genetic algorithms, especially for multi-objective optimization have been observed.

Ergonomic aspects are becoming introduced in line balancing, e.g., physical effort of operators as well as the risks that they face (Choi, 2009; Otto and Scholl, 2011; Xu et al., 2012), learning effect (Costa and Miralles, 2009; Toksarı et al., 2008, 2010), fatigue (Digiesi et al., 2009), and workers' skills (Miralles et al., 2007, 2008; Wong et al., 2006). Environmental issues have also been recently considered (Altekin and Kandiller, 2008; Gungor and Gupta, 2001; McGovern and Gupta, 2006, 2007a, 2007b, 2011). This seems to be crucial for the future research.

Even though several authors have treated the problem of rebalancing for manual flow lines (Altekin and Akkan, 2012; Gamberini et al., 2006, 2009; Grangeon et al., 2011; Yang et al., 2011), this topic appears to be still underdeveloped. Another important issue concerning the re-balancing of a flow line equipped with machines having different levels of flexibility seems even less explored. Since this problem frequently appears in industry, an effort has to be made in order to suggest appropriate mathematical models and efficient solution methods.

Line balancing should be also considered with robust optimization approaches in order to take into account the uncertainty related to problem parameters. One of the approaches used in the literature is based on scenario planning where each scenario corresponds to the assignment of plausible values to the input data. Robust analysis searches for a solution or a set of solutions that perform well across all possible scenarios. The most widely used robust criteria rely on the worst case, such as min–max, min–max regret, and min–max relative regret (Dolgui and Kovalev, 2012; Simaria et al., 2009; Xu and Xiao, 2009). However, use of these criteria often results in conservative decisions, since they are based on the most pessimistic scenario. To introduce some tolerance for risk in the decision making, new definitions can be given for robust solutions using for example the concepts of p-robustness, α-reliable min–max regret model, or lexicographic α-robustness (Xu and Xiao, 2011). Recently, a branch and bound algorithm for robust balancing of simple assembly lines with interval task processing times has been suggested by Gurevsky et al. (2012b), where a parameter $\theta \in [0,1]$ is introduced to reflect the risk aversion coefficient of a decision maker and to control the conservatism of the solution method used. A lexicographic-order based robust approach with the objective function based on the α-worst case scenario for mixed model assembly line balancing problem has been developed by Xu and Xiao (2011). These publications open very interesting and promising research paths.

Advanced line balancing problems require a lot of a priori information about the line being designed and technological processes employed. Several alternatives should be considered, more input data should be determined. This is not a simple issue in practice and a compromise between the quantity as well as the reliability of input information and the complexity of the corresponding mathematical models needs to be found.

Another area, still little explored, is stability analysis of optimal solutions obtained for line balancing problems. Some initial work on this subject has already been presented by Gurevsky et al. (2012a, 2013) and Sotskov et al. (2006) to calculate stability radius for optimal solutions. Additional work is needed in order to develop new mathematical tools able to provide decision makers not only with solutions of adequate quality but ones that are robust to perturbations in input data.

11. Conclusion

The goal of this survey was to develop a novel taxonomy of line balancing problems backed up by recent publications since 2007. In contrast to previously published state of the art papers, a larger scope was considered including assembly, disassembly and machining line balancing problems. Several more general extensions consisting of combining line balancing, process planning, line design and other related problems have been surveyed. The approach adopted was based on the following five key model elements: number of lines being balanced, task attributes, workstation attributes, problem constraints, and objective functions.

Our analysis of the recent literature has shown that a number of recent publications on multiple lines with multi-line workstations or workers performing tasks for more than one line are available. However, configurations with multiple independent lines and parallel lines with crossovers between some workstations are less well treated in the literature.

The comparison on task attributes has revealed that having multiple task attributes (not limited to task times) is common for assembly, machining and disassembly contexts. These attributes are not necessarily constant. It is noted that the number of task attributes considered has increased in recent publications and their mathematical representations have become more diversified. The use of workstation scalar- and vector- attributes has also been suggested to represent resource (equipment, workers, etc) allocation problems considered jointly with line balancing. It is concluded that the problem of resource allocation has not been considered for disassembly lines as yet. Moreover, mathematical models developed for assembly and machining lines cannot be applied directly for disassembly lines.

The examination of different constraints suggests that almost all constraint types are common for all industrial contexts studied (machining, assembly, disassembly, etc). Specific and more complex constraints are imposed rather by adding decisions often considered jointly with line balancing, as for example task scheduling, than by the manufacturing environment considered.

The survey has also analysed the criteria employed in multi-objective optimization for line balancing. This brings to light the combinations of criteria most frequently used, as well as the priorities usually assigned to them.

The solution methods developed for line balancing problems have been classified in three categories: exact methods, simple heuristics and metaheuristics. Special attention has been paid for the methods employed for multi-objective decision making.

A significant endeavour has been made in the literature in order to introduce more task attributes, evaluate more performance measures, as well as consider more constraints and objective functions. Nevertheless, a gap still exists between mathematical formulations and real life problems. A number of research perspectives have been highlighted in this paper in order to fill this gap.

We hope that this taxonomy may be useful to all of the line balancing community to understand better the state-of-the-art and in selecting adequate models for new possible line balancing problems and applications. In addition, this taxonomy can facilitate the search for similar existing problems as well as common models that can be applied in different industries (e.g., disassembly, machining, semiconductor manufacture, etc.).

Finally, the taxonomy proposed in this article is a first step and we will work on additional examples and links between different line balancing problems to further develop this approach.

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